The interchange law for horizontal composition of natural transformations

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'ct:interchange' (source lines 416-420, Lemma 9.2.8); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. No context span is retained: the statement and proof are self-contained and use no label from outside the retained range.

Editorial note. The retained counter values (chapter 9, section 2, theorem offset 7) make the counted lemma print as Lemma 9.2.8, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: the interchange law for horizontal composition (Lemma 9.2.8).

Allowed standard prerequisites (editorial).

- Whiskering and composition of natural transformations. delta G and H gamma denote the whiskered composites of a natural transformation with a functor.. They are the two sides of the interchange identity.
- Componentwise equality of natural transformations. Two natural transformations are equal as soon as their components are equal at every object.. It is the reduction performed at the start of the proof.

Lemma 9.2.8. *For functors $F, G : A \to B$ and $H, K : B \to C$ and natural transformations $\gamma : F \to G$ and $\delta : H \to K$, we have*

$$(\delta G)(H\gamma) = (K\gamma)(\delta F).$$

Proof. It suffices to check componentwise: at $a : A$ we have

$$\begin{aligned}
((\delta G)(H\gamma))_a &\equiv (\delta G)_a (H\gamma)_a \\
&\equiv \delta_{Ga} \circ H(\gamma_a) \\
&= K(\gamma_a) \circ \delta_{Fa} && \text{(by naturality of } \delta) \\
&\equiv (K\gamma)_a \circ (\delta F)_a \\
&\equiv ((K\gamma)(\delta F))_a. && \square
\end{aligned}$$